Lab 3: Architecture Justification

Problem Statement #14: Corporate Mental Wellness & Counseling Platform · Team 14 · Bhuvan Kumar P

Architecture selection: *We chose Microservices Architecture for the Corporate Mental Wellness & Counseling Platform.*

Architectural choice

The platform is split into small, independently deployable services behind an API Gateway: Identity & Pseudonymisation, Assessment, Booking & Scheduling, Wellness Records & Reporting, and Notification. Each service owns its own data. Layered architecture was rejected because one shared database would put the pseudo-ID mapping beside counselling data. Client-server was rejected because a single server is a single point of failure for crisis escalation.

Reason 1: Privacy needs hard isolation of identity

The whole platform depends on employees trusting that their employer cannot identify them (FR-001, NFR-002). With microservices, the pseudo-ID ↔ employee mapping lives only in the Identity Service and its own encrypted Identity Store. The other services see nothing but the random pseudo-ID. A leak of the Counselling Database therefore identifies nobody. A layered monolith cannot enforce this boundary at deployment level.

Reason 2: Uneven load and independent change

Booking traffic peaks sharply (Monday mornings, many timezones; NFR-001: 500 concurrent users, slot list in under 2 s), while aggregate reporting runs rarely. Microservices let us scale only the Booking & Scheduling Service. The Assessment Service and its crisis referral (FR-005) also stay available if Reporting or Notification fails, because faults are isolated per service.

Security advantage

Defence in depth with least privilege. The API Gateway terminates TLS 1.3 and validates a pseudo-ID JWT before any request reaches a service. Each service has its own credentials and encryption keys (AES-256 at rest), so compromising one service does not expose the identity mapping or the session notes. The Reporting Service enforces the ten-employee minimum before any aggregate leaves the system.

Performance benefit

Independent horizontal scaling plus asynchronous messaging. Only the Booking Service is replicated at peak, and it locks a slot in a single local transaction (FR-003). Confirmations are handed to the Notification Service over an async queue (AMQP), so a slow email or push provider never delays the booking response. This keeps the slot list and booking under the 2-second target.